National Institute of Business Management

Diploma in Software Engineering 24.2F

Project Proposal

Project Title: Database Management System for Serendib Grand Hotel

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1.Organizational Description

The "Serendib Grand Hotel" is a one-branch hotel providing three types of rooms called Standard, Deluxe and Suite and some guest services such as Room service, laundry services, Spa treatment, and Airport Transfers. Like any other hotel, this particular hotel produces large amounts of interconnected data during its daily operations, including data about room inventory, guest bookings, usage of services, payment transactions, staff and departmental administration and housekeeping. Currently this data is spread throughout various manual and non-connected spreadsheets and therefore creates problems related to duplication of data, booking clashes, poor reporting and errors.

2. Problem Description and Project Purpose

Without having a good database structure, it will be impossible to track room availability, guest profile, payment balance and workload of the hotel's staff. This project aims at designing and implementing relational database management system containing all mentioned above data areas.

3. Scope of the Project

The project involves covering the entire database development lifecycle, starting from requirement analysis to a functional physical implementation. The scope involves:

- Entities, attributes, and relationships analysis pertaining to hotel operations (at least 8 entities, including room specialization, department/staff management, booking, services, payments, and housekeeping).
- Conceptual Design - ER (Entity Relationship) diagram consisting of all entities, attributes, relationships, cardinality ratio, and participation constraints.
- Logical Design – Conversion of the ER diagram into a relational schema with primary key and foreign keys followed by normalization (1NF to BCNF).
- Physical Design – Implementation of the database in Microsoft SQL Server that involves creation of tables with constraints, insertion of at least 10 sample records per table, query for SELECT statement, GROUP BY / HAVING / ORDER BY, JOIN, and role-based user access permission.
- Documentation – Full project report, commented SQL code and presentation of design and advantages.

The scope of the project is confined to one hotel branch only and will not involve online booking, payment gateway transaction or synchronization of multiple branches, which can be considered for future enhancement.

4. Expected outcome

A normalized and properly constrained relational database model, which would effectively represent the activities of Serendib Grand Hotel, and at the same time allows for the quick extraction of information via wellconstructed queries.